Switching and Transmission:

Transmission

- It is the propagation of a signal, message or other form of intelligence by any means such as optical fiber, wire, or visual means.
- Our definition is not so broad. Transmission provides the transport of a signal from an end-user source to the destination such that the signal quality at the destination meets certain performance criteria.

Switching:

It refers to directing the signal (data) to the desired destination that the transmitted signal travels by the closing of switches in either the space domain or the time domain or some combination(s) of the two.

- Switching: selects and directs communication signals to a specific user or a group of users
- Transmission: delivers the signals in some way from source to the far-end user with an acceptable signal quality

Switching

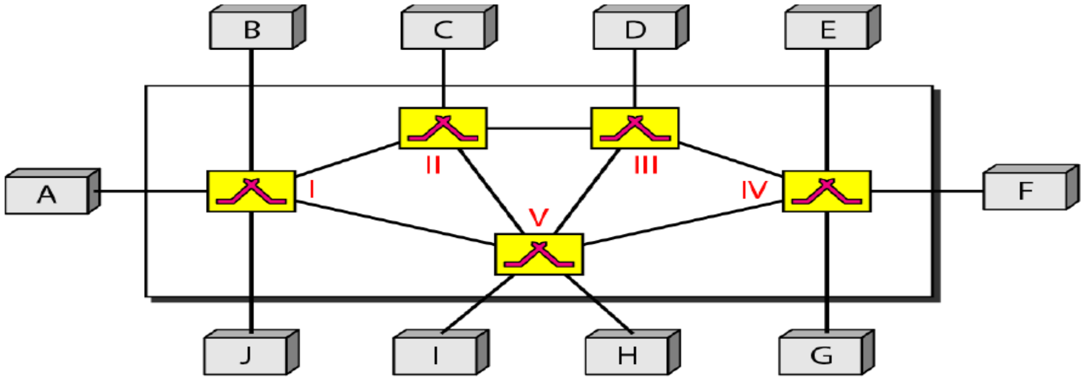
A network is a set of connected devices. Whenever we have multiple devices, we have the problem of how to connect them to make one-to-one communication possible.

One solution is to make a point-to-point connection between each pair of devices (a mesh topology) or between a central device and every other device (a star topology). These methods, however, are impractical and wasteful when applied to very large networks. The number and length of the links require too much infrastructure to be cost-efficient, and the majority of those links would be idle most of the time.

A better solution is switching

A switched network consists of a series of interlinked nodes, called switches. Switches are devices capable of creating temporary connections between two or more devices linked to the switch.

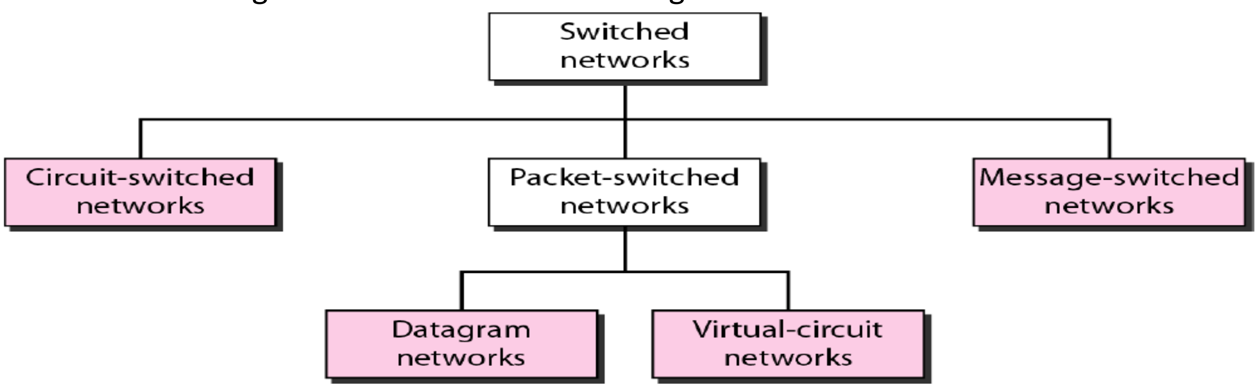
In a switched network, some of these nodes are connected to the end systems (computers or telephones). Others are used only for routing. Figure shows a switched network.



We can then divide today's networks into three broad categories:

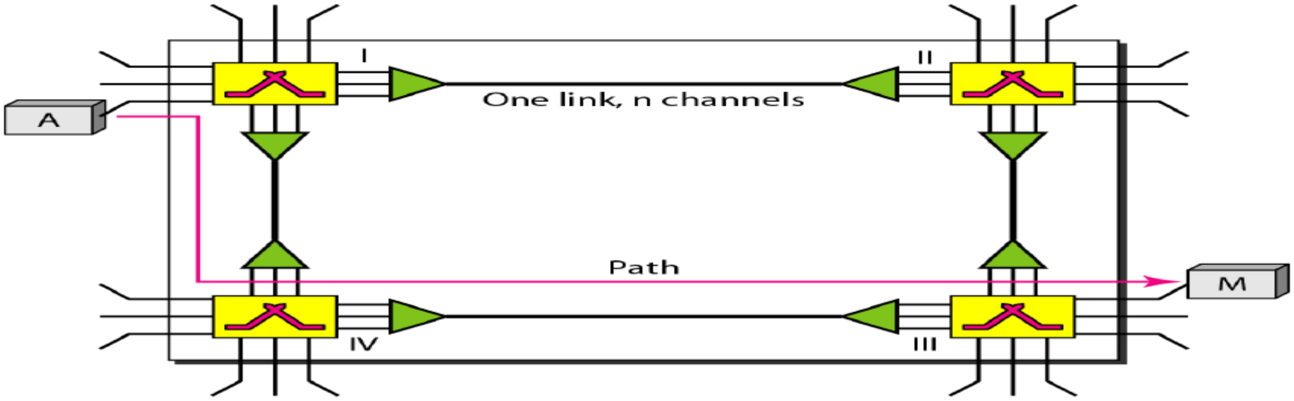
- Circuit switched networks,
- Packet-switched networks,
- Message-switched.

Packet switched networks can further be divided into two subcategories-virtual-circuit networks and datagram networks as shown in Figure.



CIRCUIT-SWITCHED NETWORKS

- A circuit-switched network consists of a set of switches connected by physical links.
- A connection between two stations is a dedicated path made of one or more links. However, each connection uses only one dedicated channel on each link.
- Each link is normally divided into n channels by using FDM or TDM.
- In circuit switching, the resources need to be reserved during the setup phase; the resources remain dedicated for the entire duration of data transfer until the teardown phase.



Three Phases

The actual communication in a circuit-switched network requires three phases: connection setup, data transfer, and connection teardown.

Setup Phase

Before the two parties (or multiple parties in a conference call) can communicate, a dedicated circuit (combination of channels in links) needs to be established. Connection setup means creating dedicated channels between the switches.

Data Transfer Phase

After the establishment of the dedicated circuit (channels), the two parties can transfer data.

Teardown Phase

When one of the parties needs to disconnect, a signal is sent to each switch to release the resources.

Features of Circuit Switching Networks

Efficiency

It can be argued that circuit-switched networks are not as efficient as the other two types of networks because resources are allocated during the entire duration of the connection. These resources are unavailable to other connections.

Delay

Although a circuit-switched network normally has low efficiency, the delay in this type of network is minimal. During data transfer the data are not delayed at each switch; the resources are allocated for the duration of the connection.

The total delay is due to the time needed to create the connection, transfer data, and disconnect the circuit.

Packet Switching Networks:

- In a packet-switched network, there is no resource reservation; resources are allocated on demand.
- The allocation is done on a first come, first-served basis. When a switch receives a packet, no matter what is the source or destination, the packet must wait if there are other packets being processed.
- This lack of reservation may create delay.

Basics of Packet Switching:

• Data transmitted in small packets — Typically 1000 octets —Longer messages split into series of packets —Each packet contains a portion of user data plus Control info —Routing (addressing) info

• Packets are received, stored briefly (buffered) and passed on to the next node —Store and forward.

• Line efficiency — Single node to node link can be shared by many packets over time — Packets queued and transmitted as fast as possible

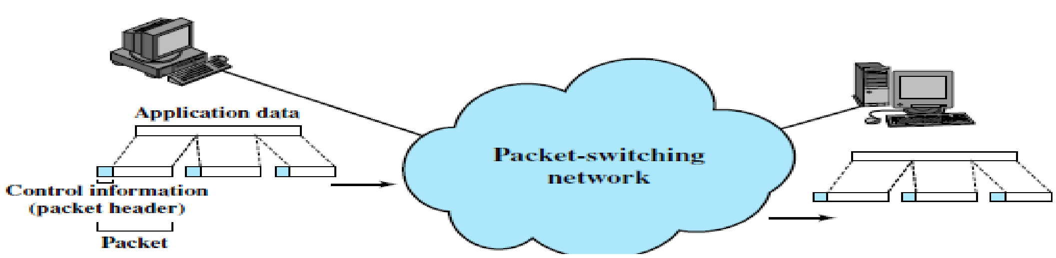
• Data rate conversion — Each station connects to the local node at its own speed — Nodes buffer data if required to equalize rates

• Packets are accepted even when network is busy — Delivery may slow down

• Priorities can be used.

Packet Switching Technique

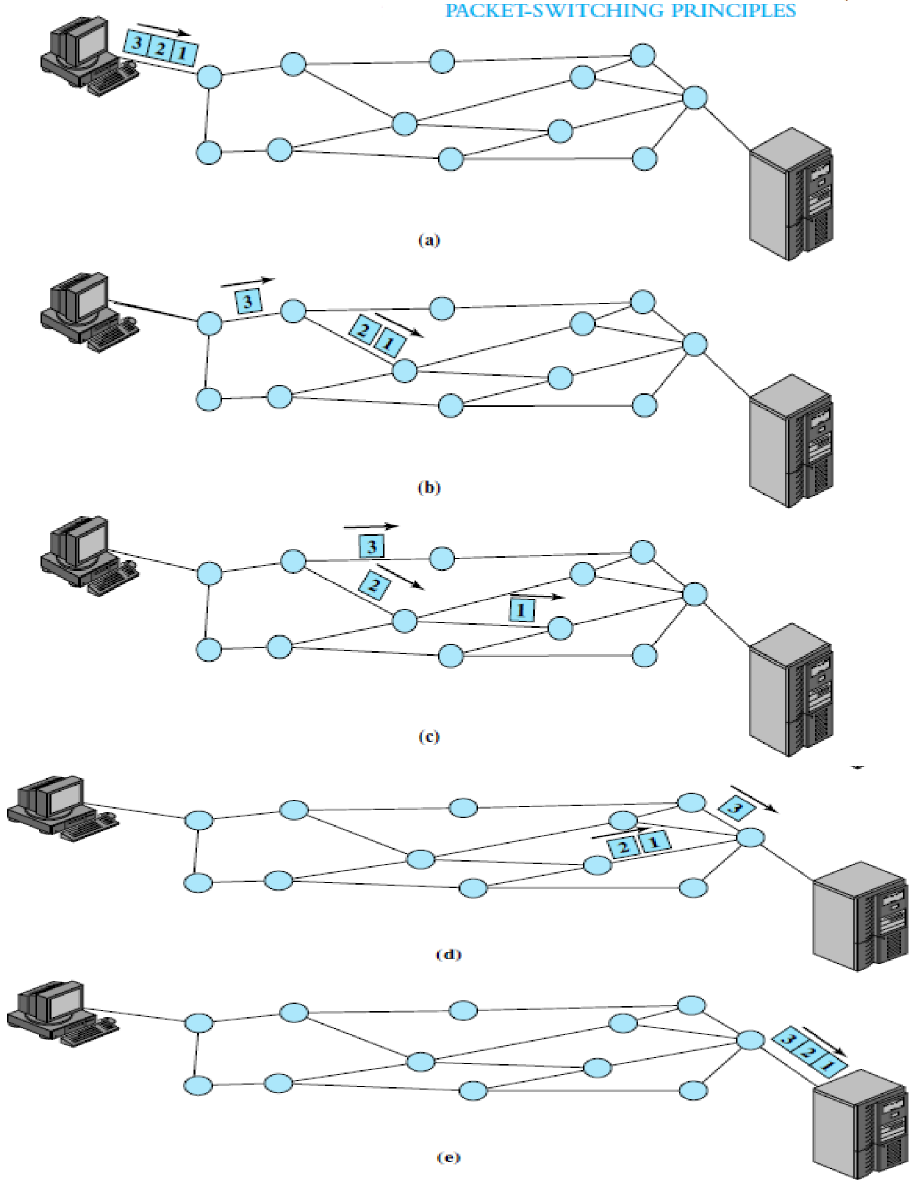
- ✓ Station breaks long message into packets
- ✓ Packets sent one at a time to the network
- ✓ Packets handled in two ways —**Datagram** —**Virtual circuit.**



In Datagram

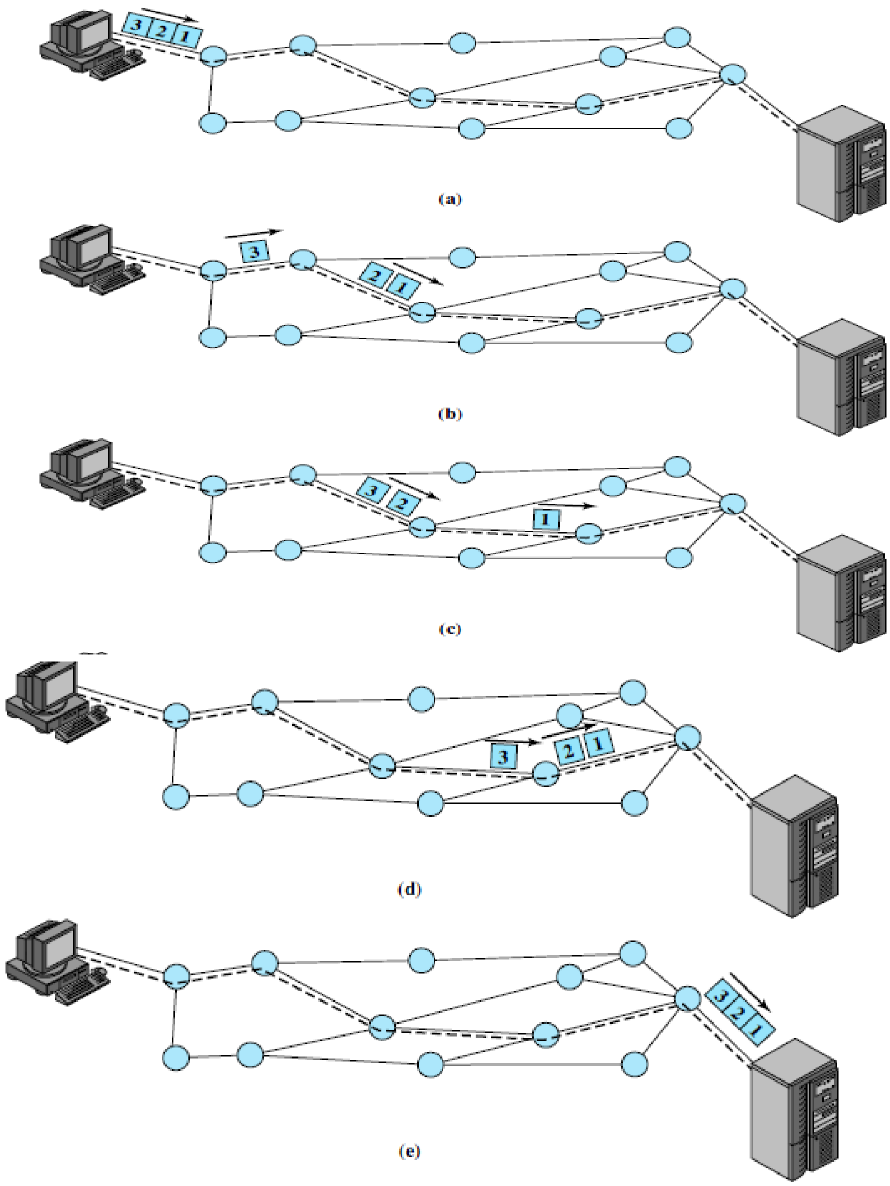
•Each packet treated independently -- Packets can take any practical route - Packets may arrive out of order --Packets may go missing - Up to receiver to re-order packets and recover from missing packets.

PACKET-SWITCHING PRINCIPLES



In Virtual Circuit

- Preplanned route established before any packets sent
- Call request and call accept packets establish connection (handshake)
- Each packet contains a virtual circuit identifier instead of destination address
- No routing decisions required for each packet • Clear request to drop circuit
- Not a dedicated path.



3 2 1
(a)
3
2 1
(b)
3 2
1
(c)
3
2 1
(d)
3 2 1
(e)

Comparison of Communication Switching Techniques

Circuit Switching	Datagram Packet Switching	Virtual Circuit Packet Switching
Dedicated transmission path	No dedicated path	No dedicated path
Continuous transmission of data	Transmission of packets	Transmission of packets
Fast enough for interactive	Fast enough for interactive	Fast enough for interactive
Messages are not stored	Packets may be stored until delivered	Packets stored until delivered
The path is established for entire conversation	Route established for each packet	Route established for entire conversation
Call setup delay; negligible transmission delay	Packet transmission delay	Call setup delay; packet transmission delay
Busy signal if called party busy	Sender may be notified if packet not delivered	Sender notified of connection denial
Overload may block call setup; no delay for established calls	Overload increases packet delay	Overload may block call setup; increases packet delay
Electromechanical or computerized switching nodes	Small switching nodes	Small switching nodes
User responsible for message loss protection	Network may be responsible for individual packets	Network may be responsible for packet sequences
Usually no speed or code conversion	Speed and code conversion	Speed and code conversion
Fixed bandwidth	Dynamic use of bandwidth	Dynamic use of bandwidth
No overhead bits after call setup	Overhead bits in each packet	Overhead bits in each packet

Network Structure

When you classify computer networks from a structural (architectural) perspective, a very common and important division is into:

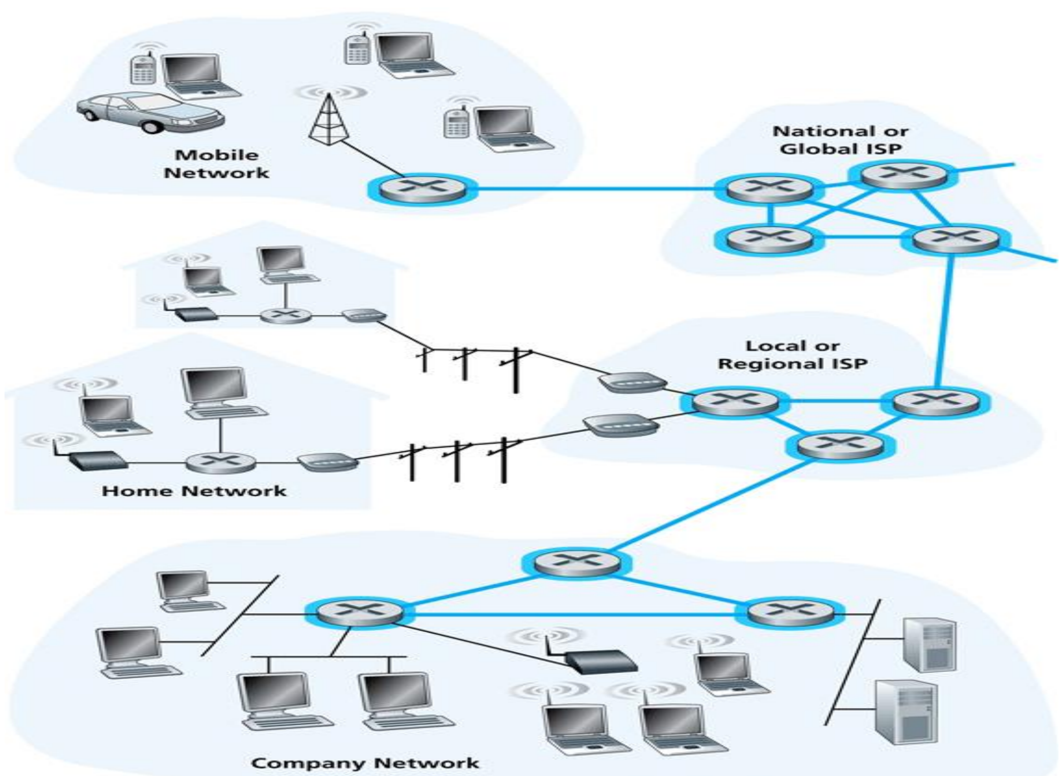
1) Network Core (Network Code)

- This is the **backbone** of the network.
- It is used as **interconnecting** devices
- It consists of **high-capacity routers and switches** that carry large volumes of data.

2) Network Edge (Access Network)

This is where **end users connect** to the network. It acts as **“entry and exit points”** of the network.

- Hosts: clients – users   examples : Phone –PC – Laptop etc.....
- Servers often in DC



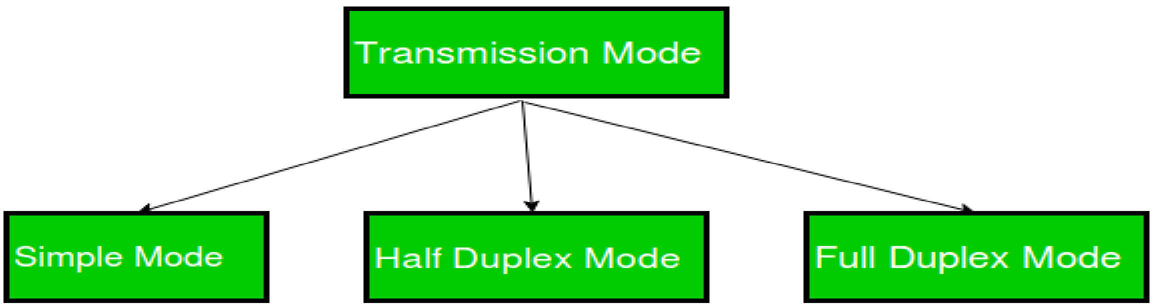
Transmission Modes in Computer Networks

Networks can be classified based on different criteria; one of them is according to Data Flow or Flow Direction.

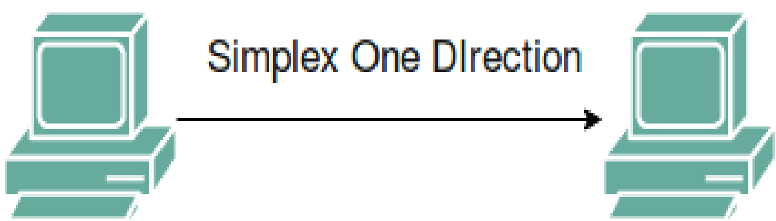
Transmission mode means transferring data between two devices. It is also known as a communication mode. Buses and networks are designed to allow communication to occur between individual devices that are interconnected.

Communication between two devices can be simplex, half-duplex, or full-duplex as shown in figure below

There are three types of transmission mode:-



Simplex:



- ✓ In simplex mode, the communication is unidirectional, as on a one-way street.
- ✓ Only one of the two devices on a link can transmit; the other can only receive.
- ✓ Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output.
- ✓ The simplex mode can use the entire capacity of the channel to send data in one direction.

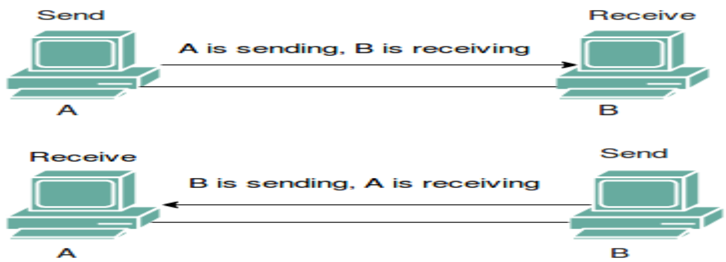
Advantages:

- Simplex mode is the easiest and most reliable mode of communication.
- It is the most cost-effective mode, as it only requires one communication channel.
- There is no need for coordination between the transmitting and receiving devices, which simplifies the communication process.
- Simplex mode is particularly useful in situations where feedback or response is not required, such as broadcasting or surveillance.

Disadvantages:

- Only one-way communication is possible.
- There is no way to verify if the transmitted data has been received correctly.
- Simplex mode is not suitable for applications that require bidirectional communication.

Half-Duplex:



- ✓ In half-duplex mode, each station can both transmit and receive, but not at the same time.
- ✓ When one device is sending, the other can only receive, and vice versa.
- ✓ The half-duplex mode is like a one-lane road with traffic allowed in both directions.

Advantages:

- Half-duplex mode allows for bidirectional communication, which is useful in situations where devices need to send and receive data.
- It is a more efficient mode of communication than simplex mode, as the channel can be used for both transmission and reception.
- Half-duplex mode is less expensive than full-duplex mode, as it only requires one communication channel.

Disadvantages:

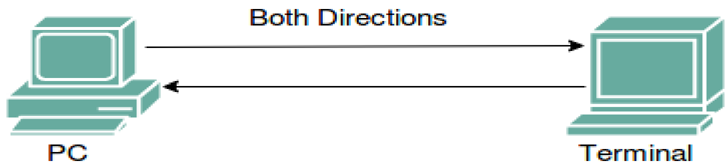
- Half-duplex mode is less reliable than Full-Duplex mode, as both devices cannot transmit at the same time.
- There is a delay between transmission and reception, which can cause problems in some applications.
- There is a need for coordination between the transmitting and receiving devices, which can complicate the communication process.

Full-Duplex:

In full-duplex mode, both stations can transmit and receive simultaneously. In full duplex mode, signals going in one direction share the capacity of the link with signals going in another direction, this sharing can occur in two ways:

- ✓ Either the link must contain two physically separate transmission paths, one for sending and the other for receiving.
- ✓ Or the capacity is divided between signals traveling in both directions.

Full-duplex mode is used when communication in both directions is required all the time. The capacity of the channel, however, must be divided between the two directions.
Example: Telephone Network in which there is communication between two persons by a telephone line, through which both can talk and listen at the same time.



Advantages:

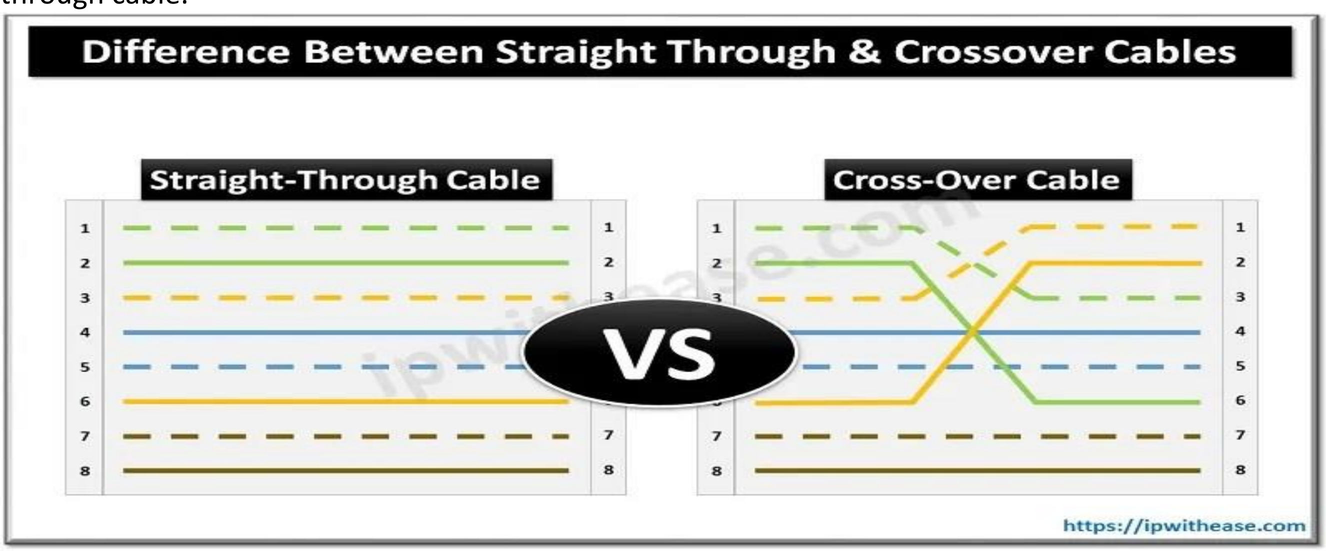
- Full-duplex mode allows for simultaneous bidirectional communication, which is ideal for real-time applications such as video conferencing or online gaming.
- It is the most efficient mode of communication, as both devices can transmit and receive data simultaneously.
- Full-duplex mode provides a high level of reliability and accuracy, as there is no need for error correction mechanisms.

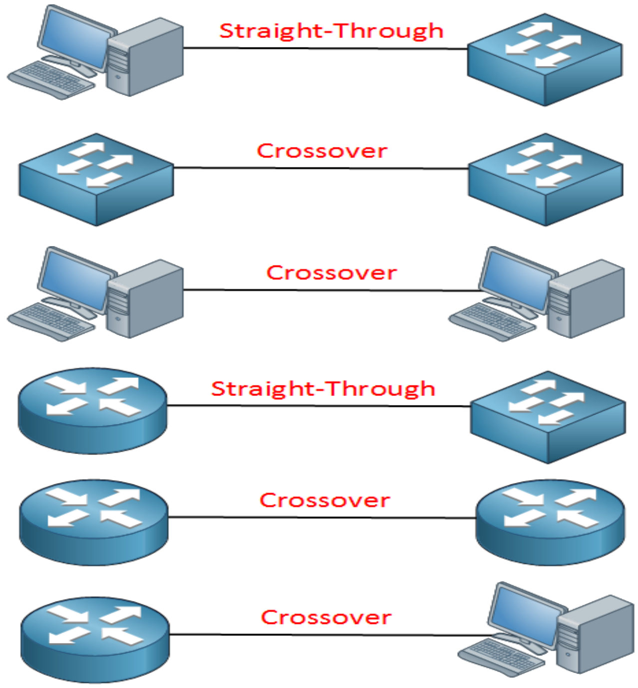
Disadvantages:

- Full-duplex mode is the most expensive mode, as it requires two communication channels.
- It is more complex than simplex and half-duplex modes, as it requires two physically separate transmission paths or a division of channel capacity.
- Full-duplex mode may not be suitable for all applications, as it requires a high level of bandwidth and may not be necessary for some types of communication.

In Cisco Technology :

The cable shown in Figure below is called a ***straight-through cable*.** A straight-through cable connects pin 1 on one end of the cable with pin 1 on the other end, pin 2 on one end to pin 2 on the other, and so on. If you hold the cable so that you compare both connectors side by side, with the same orientation for each connector, you should see the same color wires for each pin with a straight-through cable.

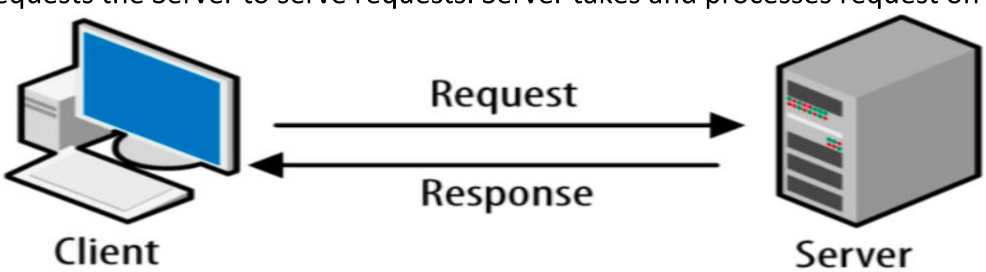




Network Architecture

Computer networks can be discriminated into various types such as Client-Server, peer-to-peer or hybrid, depending upon its architecture.

- ✓ **In Client-Server,** There can be one or more systems acting as Server. Other being Client, requests the Server to serve requests. Server takes and processes request on behalf of Clients.



- ✓ **In peer-to-peer,** Two systems can be connected Point-to-Point, or in back-to-back fashion. They both reside at the same level and called peers.

- ✓ There can be **Hybrid network** which involves network architecture of both the above types.

Network Criteria

A network must be able to meet a certain number of criteria. The most important of these are performance, reliability, and security.

Performance:

Performance can be measured in many ways, including **transit time and response time**.

- ✓ Transit time is the amount of time required for a message to travel from one device to another.
- ✓ Response time is the elapsed time between an inquiry and a response.
- ✓ The performance of a network depends on a number of factors, including the number of users, the type of transmission medium, the capabilities of the connected hardware, and the efficiency of the software.
- ✓ Performance is often evaluated by two networking metrics: throughput and delay.
- ✓ We often need more throughputs and less delay.

Reliability:

In addition to accuracy of delivery, network reliability is measured by:

- ✓ The frequency of failure, the time it takes a link to recover from a failure
- ✓ Recovery time

Security:

Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Classification of Networks based on connection type (Line Configuration)

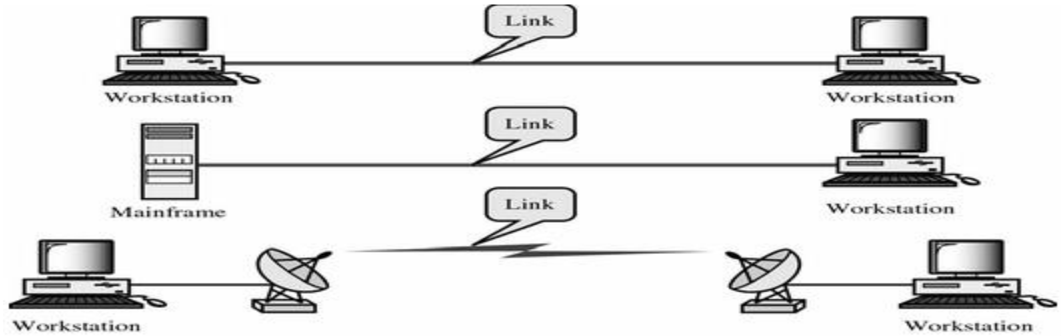
Line configuration refers to the way two or more communication devices attached to a link. Line configuration is also referred to as connection. A Link is the physical communication pathway that transfers data from one device to another. For communication to occur, two devices must be connected in same way to the same link at the same time.

There are two possible line configurations:

1. Point-to-Point. 2. Multipoint.

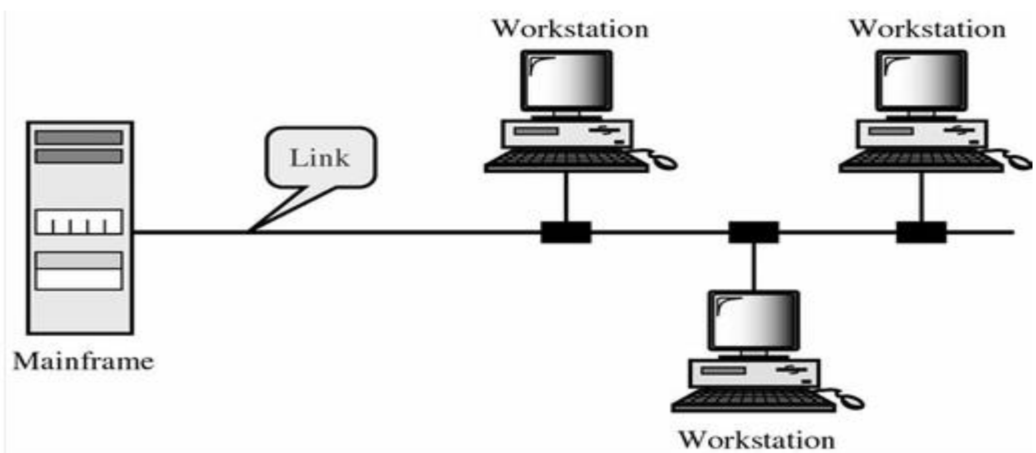
Point-to-Point

- ✓ A point-to-point connection provides a dedicated link between two devices.
- ✓ The entire capacity of the link is reserved for transmission between those two devices.



Multipoint (also called Multidrop)

- ✓ The connection is one in which more than two specific devices share a single link.
- ✓ In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.
- ✓ If several devices can use the link simultaneously, it is a *spatially shared* connection. If users must take turns, it is a *timeshared* connection.



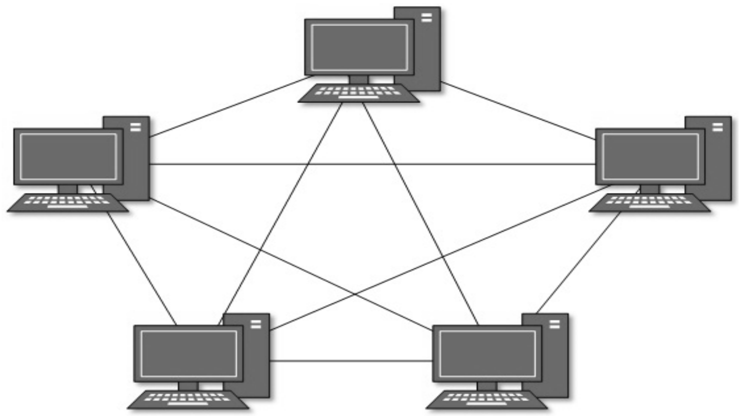
Physical Topology of networks:

The term ***physical topology*** refers to the way in which a network is laid out physically. One or more devices connect to a link; two or more links form a topology. The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another. There are four basic topologies possible: mesh, star, bus, and ring.

Mesh Topology

- ✓ In a mesh topology, every device has a dedicated point-to-point link to every other device.
- ✓ The term *dedicated* means that the link carries traffic only between the two devices it connects
- ✓ To find the number of physical links in a fully connected mesh network with *n* nodes,

We need $n(n - 1)/2$ physical links.



Advantages:

1. The use of dedicated links guarantees that each connection can carry its own data load, thus **eliminating the traffic problems** that can occur when links must be shared by multiple devices.

2. A **mesh topology is robust.** If one link becomes unusable, it does not incapacitate the entire system. Third,
3. **Privacy and Security**. When every message travels along a dedicated line, only the intended recipient sees it. Physical boundaries prevent other users from gaining access to messages.
4. Finally, point-to-point links make fault identification and **fault isolation easy.** Traffic can be routed to avoid links with suspected problems. This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

Disadvantages:

1. Disadvantage of a mesh is related to the **amount of cabling** because every device must be connected to every other device, **installation and reconnection are difficult.**

2. Second, the sheer **bulk of the wiring** can be greater than the available space (in walls, ceilings, or floors) can accommodate.
3. Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

Star Topology:

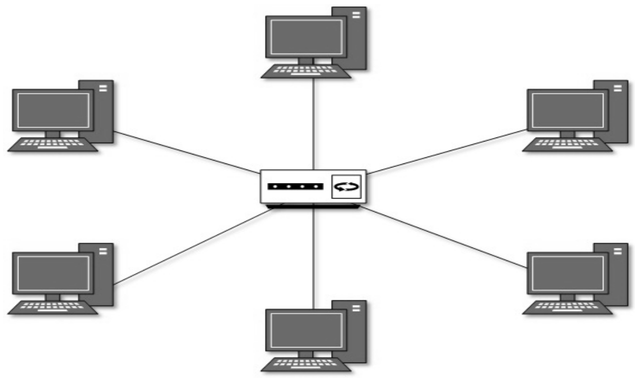
- ✓ In a star topology, each device has a dedicated point-to-point link only to a central controller. The Centralized device can be any of the following:
  - Layer-1 device such as hub or repeater
  - Layer-2 device such as switch or bridge
  - Layer-3 device such as router or gateway
- ✓ The devices are not directly linked to one another. Unlike a mesh topology, a star topology does not allow direct traffic between devices.
- ✓ The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.

Advantages:

1. A star topology is **less expensive** than a mesh topology. In a star, each device needs only one link.

2. It is **easy to install and reconfigure**. Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.

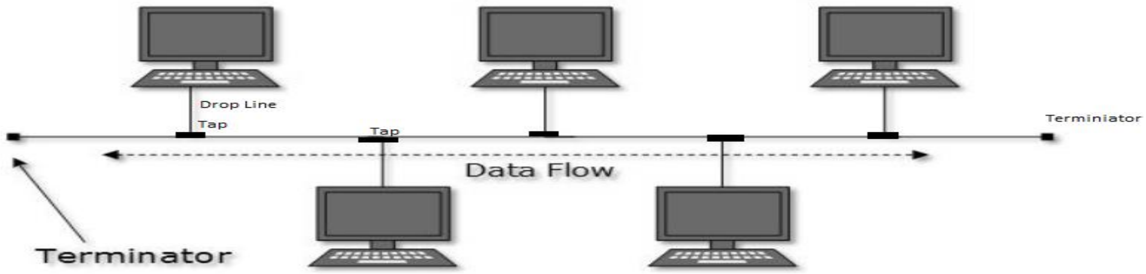
3. Other advantages include **robustness**. If one link fails, only that link is affected. All other links remain active.



One big disadvantage of a star topology is the dependency of the whole topology on **one single point**, the hub. If the hub goes down, the whole system is dead. Although a star requires far less cable than a mesh, each node must be linked to a central hub.

Bus Topology:

- ✓ The preceding examples all describe point-to-point connections. A **bus topology,** on the other hand, is multipoint.
- ✓ One long cable acts as a **backbone** to link all the devices in a network.
- ✓ Bus topology may have problem while multiple hosts sending data at the same time. Therefore, Bus topology either uses CSMA/CD technology.
- ✓ The data is sent in only one direction and as soon as it reaches the extreme end, the **terminator** removes the data from the line
- ✓ Devices are connected to the bus cable by drop lines and taps.
    - A drop line is a connection running between the device and the main cable.
    - A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.
- ✓ As a signal travels along the backbone, therefore, it becomes weaker and weaker as it travels farther and farther. For this reason there is a limit on the number of taps a bus can support and on the distance between those taps.



Advantages

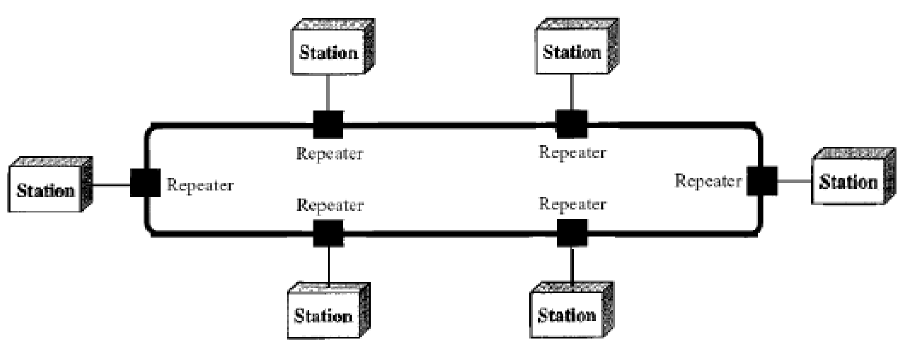
It includes **ease of installation**. Backbone cable can be laid along the most efficient path, then connected to the nodes by drop lines of various lengths. In this way, a bus uses less cabling than mesh or star topologies Only the backbone cable stretches through the entire facility. Each drop line has to reach only as far as the nearest point on the backbone.

Disadvantages

- ✓ It includes **difficult reconnection and fault isolation**. A bus is usually designed to be optimally efficient at installation. It can therefore be difficult to add new devices. Signal reflection at the taps can cause degradation in quality.
- ✓ In addition, **a fault or break in the bus cable stops all transmission**, even between devices on the same side of the problem. The damaged area reflects signals back in the direction of origin, creating noise in both directions.

Ring Topology

- ✓ Each device has a dedicated point-to-point connection with only the two neighboring devices on either side of it.
- ✓ A signal is passed along the ring in one direction, from device to device, until it reaches its destination.
- ✓ In a ring network, packets of data travel from one device to the next until they reach their destination. Most ring topologies allow packets to travel only in one direction, called a unidirectional ring network (However, **unidirectional** traffic can be a **disadvantage)**. Others permit data to move in either direction, called **bidirectional**.
- ✓ Each device in the ring incorporates a repeater. When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.
- ✓ Failure of any host results in failure of the whole ring. Thus, every connection in the ring is a point of failure.



Advantages of ring topology

A ring is relatively easy to install and reconfigure. Each device is linked to only its immediate neighbors (either physically or logically). To add or delete a device requires changing only two connections.

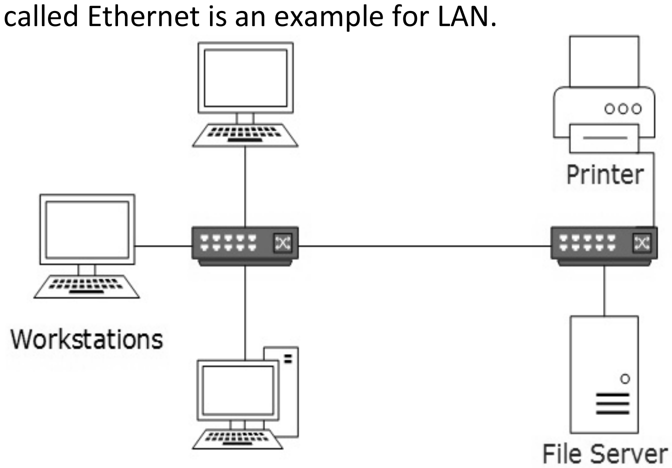
Disadvantages of ring topology

In a simple ring, a break in the ring (such as a disabled station) can disable the entire network. This weakness can be solved by using a dual ring or a switch capable of closing off the break.

Classification OF networks according to Size

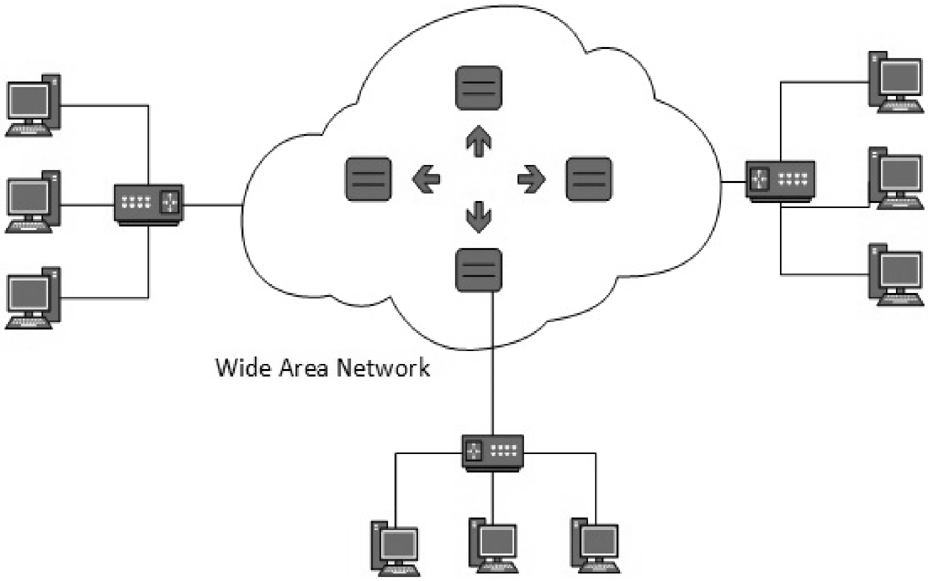
1. Local Area Network (LAN)

- ✓ A computer network spanned inside a building and operated under single administrative system is generally termed as Local Area Network (LAN).
- ✓ Number of devices and internetworking devices connected in LAN is restricted.
- ✓ Traditional LANs run at speeds of 10 Mbps to 100 Mbps.
- ✓ IEEE 802.3, popularly called Ethernet is an example for LAN.



2. Wide Area Network (WAN)

- ✓ It spans a large geographical area, often a city, country or continent.
- ✓ Generally, telecommunication networks are Wide Area Network; these networks provide connectivity to LANs
- ✓ Number of devices and internetworking devices connected in WAN are very large.
- ✓ WAN may use advanced technologies such as Asynchronous Transfer Mode (ATM), Frame Relay, and Synchronous Optical Network (SONET).
- ✓ WAN may be managed by multiple administrations.



Sometimes, Networks can be classified by their scale as follows:

Scale	Type
Vicinity	PAN (Personal Area Network) »
Building	LAN (Local Area Network) »
City	MAN (Metropolitan Area Network) »
Country	WAN (Wide Area Network) »
Planet	The Internet (network of all networks)